

Running Simulation Tests

TESTS :

1. "test_smoke"
2. "test_basic"
3. "test_dual_agent"
4. "test_dual_constrained"
5. "test_directed_channel_A"
6. "test_directed_channel_B"
7. "test_constrained"
8. "test_randomized"
9. "test_randomized_stress"
10. "test_reset_randomized"
11. "test_cross_closure"

To run all tests and generate coverage data execute run_regression.sh from the /tools directory.

```
./tools/run_regression.sh
```

Running single test

Method 1 (recommended for cocotb ≥ 2.0 This test environment uses cocotb 2.01) — Use COCOTB_TEST_MODULES

```
make SIM=verilator COCOTB_TEST_MODULES=sim.tests.<test name>
```

Example 1 - run **only** test_dual_agent:

Bash Terminal

```
make SIM=verilator COCOTB_TEST_MODULES=sim.tests.test_dual_agent
```

Example 2 - run **only** test_smoke:

Bash Terminal

```
make SIM=verilator COCOTB_TEST_MODULES=sim.tests.test_smoke
```

This is the cleanest and most modern way. It bypasses the regression manager and loads only the test you specify.

Method 2 — Use COCOTB_TESTCASE (older style, still works)

Run only dual_agent inside sim.tests.test_dual_agent:

Bash Terminal

```
make SIM=verilator COCOTB_TEST_MODULES=sim.tests.test_dual_agent COCOTB_TESTCASE=dual_agent
```

Run only smoke inside sim.tests.test_smoke:

Bash Terminal

```
make SIM=verilator COCOTB_TEST_MODULES=sim.tests.test_smoke COCOTB_TESTCASE=smoke
```

Method 3 — Persistent - Export environment variables

If you want to run the same test repeatedly:

Bash Terminal

```
export COCOTB_TEST_MODULES=sim.tests.test_dual_agent
export COCOTB_TESTCASE=dual_agent
make SIM=verilator
```